



DESIGN PROJECT FOUNDATIONS & CURRICULAR FRAMEWORK (SECTIONS 1.1 – 1.6)

Process Manual Alignment: Description, Significance, Objectives, Outcomes, Complex Problems & Curricular Requirements

Project: Alliance Smart Campus Digital Twin	Faculty Mentor: Dr. Ganga Holi (ASAC CSE)
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1.1 Description of Design Project

The Design Project is a culminating engineering experience in the ASAC curriculum where students apply theoretical knowledge, technical competencies, and engineering principles to solve an authentic, open-ended real-world challenge. The project spans two semesters and replicates a high-standard industrial R&D environment. For our project, **Alliance University Smart Campus Digital Twin**, the team addresses macro-scale cyber-physical inefficiencies across the 60-acre Anekal campus by unifying 3D procedural spatial graphics, IoT edge telemetry, renewable microgrid dispatch (480 kWp solar + 1.2 MWh BESS), and automated space utilization analytics.

- **Key Characteristics:** Open-ended problem statement (no single textbook answer); multi-disciplinary synthesis (graphics, power systems, embedded IoT, distributed software); structured design thinking methodology; iterative prototype fabrication and cloud deployment.

1.2 Significance in Engineering Education

- **Bridges Theory & Practice:** Connects core engineering theory (differential equations for thermal mass, trigonometric solar curves, computer graphics shader mathematics) directly to operational university infrastructure.
- **Prepares for Industry:** Replicates workplace dynamics including asynchronous Git collaboration, strict sub-100ms latency budgets, commercial tariff cost optimization, and multi-tier system integration.
- **Demonstrates Competency:** Delivers a verified public portfolio asset live on Vercel with open-source reproducibility.
- **Fulfills Accreditation (ABET / NBA):** Satisfies ABET Criterion 5 and NBA Program Outcomes (PO1 to PO12) by demanding comprehensive system design under realistic economic, environmental, social, and sustainability constraints.
- **Fosters Professional Identity:** Transitions student engineers into accountable technical owners responsible for design trade-offs.

1.3 Educational & Engineering Objectives

- In conformance with Section 1.3 of the Process Manual, the eight core educational objectives enable the student team to:
1. Apply the structured engineering Design Thinking process from field observations to cloud deployment.
 2. Incorporate multiple realistic constraints (BESCOM commercial tariffs, battery degradation costs, WebGL memory budgets).
 3. Integrate multi-disciplinary knowledge across computer science, electrical microgrids, and spatial architecture.
 4. Function effectively in a 4-member specialized engineering team with designated subsystem leads.
 5. Communicate technical specifications clearly through IEEE manuscripts, viva defense guides, and interactive dashboards.
 6. Manage project milestones rigorously using weekly sprint logs, risk registers, and review diaries.
 7. Prototype and empirically benchmark solutions against quantitative performance thresholds (60 FPS, 78.6 ms latency).
 8. Evaluate broader societal, environmental decarbonization, and campus sustainability impacts.

1.4 Project Outcomes (Tripartite Categorization)

Following the explicit outcome structure mandated in Section 1.4 of the Process Manual, the project delivers measurable achievements across three domains:

Outcome Category	Process Manual Mandated Competency	Empirical Project Implementation & Deliverable
Cognitive Outcomes	- Formulate clear problem statements from vague needs. - Evaluate alternative solutions via decision matrices. - Perform rigorous engineering analysis. - Select appropriate materials and components.	- Formulated mathematical optimization objective minimizing grid cost subject to thermal and battery constraints. - Evaluated 4 architecture patterns using a 5-criterion Pugh decision matrix. - Derived trigonometric solar curves and building thermal differential equations. - Selected React 19, Three.js r170, Blender 4.3 LTS, and ESP32 telemetry nodes.
Practical & Technical Outcomes	- Build functional prototype or simulation model. - Employ engineering software in design workflow. - Conduct empirical tests against specifications.	- Deployed live, interactive 3D WebGL digital twin on Vercel. - Scripted procedural headless Blender Python generators exporting <15 MB GLB models. - Benchmarked 60.0 FPS graphics, 78.6 ms telemetry sync, and 18.4% grid peak shaving.
Professional & Soft Outcomes	- Produce final design reports & professional presentation. - Demonstrate effective collaborative teamwork. - Manage project timeline within resources. - Reflect on ethical and environmental implications.	- Authored 21-page IEEE Term Paper Report and 22-slide ASAC presentation deck. - Maintained 12 sequential weekly meeting logs with assigned member responsibilities. - Delivered all Semester V milestones on schedule with zero budget overhead. - Embedded anonymous headcount privacy protections and carbon-reduction audits.

1.5 Complex Engineering Problems Analysis (TABLE FORM — Section 1.5)

Per Section 1.5 of the Process Manual, engineering design projects must address **Complex Engineering Problems** as defined by ABET/NBA. The following table maps every mandated attribute directly to our Smart Campus Digital Twin implementation:

Complex Problem Attribute	Process Manual Criteria	Concrete Implementation in Alliance Smart Campus Digital Twin
Deep Technical Knowledge Required	Cannot be solved by simple textbook formulas; requires synthesis of multiple engineering disciplines.	Synthesizes 3D computer graphics (WebGL custom shaders, GLTF vertex buffers), distributed IoT networks (WebSockets, Zustand store hydration), electrical power systems (microgrid power balance, BESS electrochemical SOC constraints), and thermodynamics.
Open-Ended, Multiple Solutions	No single correct answer; engineering trade-offs exist between cost, performance, safety, and aesthetics.	Navigated complex trade-offs between 3D polygonal fidelity (LOD) vs. client browser frame-rate (60 FPS), and between aggressive battery peak-shaving dispatch vs. Lithium-Iron-Phosphate (LFP) cycle degradation life.
Involves Conflicting Constraints	Lightweight vs. strong, low-cost vs. durable, fast vs. energy efficient.	High photorealistic spatial detail vs. strict web binary size (<15 MB); aggressive commercial peak shaving vs. maintaining academic occupant thermal comfort (24.0 ± 1.0 °C); streaming 1,024 sensor feeds vs. client network bandwidth.
Not Fully Defined in Advance	Requirements evolve as the team discovers technical or customer constraints.	Initial scope assumed simple energy telemetry; field investigations revealed an empirical 28.4% classroom ghost-booking rate and severe BESCOM maximum demand surcharges, expanding requirements to include automated space intelligence.
Requires Information Gathering	Literature review, vendor data, international standards (IEEE, ISO), user feedback.	Systematic synthesis of 12 peer-reviewed IEEE/Elsevier research publications (2024–2026), BESCOM commercial tariff rate cards, Alliance University Anekal campus GIS shapefiles, and direct interviews with campus facility engineers.
Non-Standard Solution Path	Cannot be solved by routine methods; requires creativity, iteration, and engineering judgment.	Broke away from traditional tabular SCADA systems by constructing a novel bidirectional cyber-physical coupling: spatial occupant density dynamically drives chiller setpoints and battery routing in a zero-install browser canvas.
Involves Real-World Constraints	Budget limits, safety regulations, supply chain issues, environmental impact.	Zero-dollar commercial software budget (100% open-source stack); non-intrusive student privacy regulations (strictly anonymous headcount data, zero biometric surveillance); adherence to Karnataka Grid Code standards.
Requires Teamwork & Coordination	Different sub-systems must interface correctly; complex integration challenges arise.	Coordinated across 4 distinct modules: 3D Graphics Architecture (Likith S), Energy Analytics (Vinodkumar), IoT Telemetry (Nikhil kumar V), and Space Intelligence (Kalyan kumar T) via unified TypeScript interfaces.

1.6 Curricular Requirements & Expected Academic Outcomes

As outlined in Section 1.6 of the Process Manual, the Design Project is structured across two distinct curricular phases:

- Design Project – I (5CS1990 / E1CSA313) [Semester V]:** Covers requirements identification, literature synthesis, system analysis, cyber-physical architecture design, and functional prototype implementation. (Achieved 100% with empirical benchmarks).
- Design Project – II (6CS1991 / E1CSA314) [Semester VI]:** Covers extended implementation, physical hardware deployment across campus buildings, closed-loop physical actuation, and multi-seasonal energy validation.
- Mandated Academic Outcomes:** (1) High-fidelity working prototype deployed live, (2) Scopus-indexed IEEE Conference Paper manuscript targeted for 2026 IEEE ICSGCE / ICAC, (3) Formal Review Diary with 12+ weekly meeting logs, and (4) Originality compliance (<8% similarity).